BOUNDEDNESS OF THE QUASI-COHERENT COHERATOR ON A QUASI-COMPACT QUASI-SEPARATED SCHEME

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- On an affine scheme the coherator functor agrees with taking the quasi-coherent sheaf associated to global derived sections, by the affine comparison and affine coherator lemmas. invoked at proof lines 5189-5193. Gives the affine case with N = 0.
- The induction principle for quasi-compact opens. invoked at proof lines 5185-5186. Organizes the reduction from U, V, and U cap V to X.
- The pushforwards along the quasi-compact open immersions have finite cohomological dimension on D_QCoh, by the quasi-coherence direct image lemma. invoked at proof lines 5201-5204. Bounds the extra cohomological spread introduced by pushing the pieces back to X.
- The coherator Mayer-Vietoris distinguished triangle of the preceding remark. invoked at proof lines 5210-5212. Assembles the vanishing on the pieces into vanishing on X.

Source reference. Original lemma 08DR: sources/stacks-chow/tags.txt, source lines 10646–10646. Cited at proof lines 5185-5186 to organize the induction; the affine base case and gluing step are inside the counted span.

Source reference. Original lemma 0CR0: sources/stacks-perfect/pinned-original-source.tex, source lines 5045–5045. Cited at proof lines 5187-5188 as the functor whose boundedness is asserted; the affine case and induction are inside the counted span.

Source reference. Original lemma 06Z0: sources/stacks-perfect/pinned-original-source.tex, source lines 454–454. Cited at proof lines 5192-5193 for the affine description of DQ_X; the affine case conclusion is inside the counted span.

Source reference. Original lemma 08D9: sources/stacks-perfect/pinned-original-source.tex, source lines 1670–1670. Cited at proof lines 5192-5193 together with the comparison lemma; the affine case conclusion is inside the counted span.

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Source reference. Original lemma 08D5: sources/stacks-perfect/pinned-original-source.tex, source lines 661–661. Cited at proof lines 5201-5204 for the finite cohomological dimension of the three pushforwards; the uniform bound N is inside the counted span.

Source reference. Original remark 0CR2: sources/stacks-perfect/pinned-original-source.tex, source lines 5156–5156. Cited at proof lines 5210-5212 to assemble the vanishing on the pieces; the conclusion of the lemma is inside the counted span.

Lemma 0.2. *Let X be a quasi-compact and quasi-separated scheme. The functor DQ_X of Lemma 0CR0 has the following boundedness property: there exists an integer $N = N(X)$ such that, if K in $D(\mathcal{O}_X)$ with $H^i(U, K) = 0$ for U affine open in X and $i \notin [a, b]$, then the cohomology sheaves $H^i(DQ_X(K))$ are zero for $i \notin [a, b + N]$.*

Proof. We will prove this using the induction principle of Cohomology of Schemes, Lemma 08DR.

If X is affine, then the lemma is true with $N = 0$ because then $RQ_X = DQ_X$ is given by taking the complex of quasi-coherent sheaves associated to $R\Gamma(X, K)$. See Lemmas 06Z0 and 08D9.

Assume U, V are quasi-compact open in X and the lemma holds for U, V, and $U \cap V$. Say with integers $N(U)$, $N(V)$, and $N(U \cap V)$. Now suppose K is in $D(\mathcal{O}_X)$ with $H^i(W, K) = 0$ for all affine open $W \subset X$ and all $i \notin [a, b]$. Then $K|_U$, $K|_V$, $K|_{U \cap V}$ have the same property. Hence we see that $RQ_U(K|_U)$ and $RQ_V(K|_V)$ and $RQ_{U \cap V}(K|_{U \cap V})$ have vanishing cohomology sheaves outside the interval $[a, b + \max(N(U), N(V), N(U \cap V))$. Since the functors $Rj_{U,*}$, $Rj_{V,*}$, $Rj_{U \cap V,*}$ have finite cohomological dimension on D_{QCoh} by Lemma 08D5 we see that there exists an N such that $Rj_{U,*}DQ_U(K|_U)$, $Rj_{V,*}DQ_V(K|_V)$, and $Rj_{U \cap V,*}DQ_{U \cap V}(K|_{U \cap V})$ have vanishing cohomology sheaves outside the interval $[a, b + N]$. Then finally we conclude by the distinguished triangle of Remark 0CR2. □